

Topgyal Gurung

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Full Stack Software Engineer with strong backend and database expertise, supported by a Bachelor's degree in Computer Science and a solid foundation in Mathematics. Experienced in designing scalable systems and deploying full-stack applications across diverse environments. Skilled in JavaScript, TypeScript, Python, Java, SQL, and modern frameworks, with hands-on experience in backend APIs, data engineering, and security-focused development. Adept at translating complex requirements into reliable, high-impact solutions, while continuing to strengthen cloud and AI skills including AWS, BigQuery, and LLM integrations.

Technical Skills

PROGRAMMING & LANGUAGES: JavaScript, TypeScript, Python, Java, SQL, HTML, CSS, Shell Scripts
FRAMEWORKS & LIBRARIES: React, Redux, Next.js, Node.js, Express, Spring Boot, Docker, Kubernetes
DATABASES & CLOUD: PostgreSQL, MongoDB, Redis, GCP BigQuery, AWS (Developer Associate in progress)
APIs & TOOLS: REST, GraphQL, Google Maps, Cloudinary, Stripe, Git, GitHub Actions (CI/CD), TDD, Vim
AI & EMERGING TECH: Vercel AI SDK, LangChain, OpenAI APIs, Claude APIs, LLM application development
CERTIFICATIONS: Google AI Essentials, AWS Academy Cloud Foundations, Python for Everybody, Google Data Analytics Professional Certificate, Claude Code in Action (Anthropic)

Project Highlights

Himalayan Connect NYC | 01/2025 - Present

- Engineered and deployed a full-stack community platform using Next.js, TypeScript, Prisma, and PostgreSQL, translating Figma designs into a production-ready application for discovering local community resources.
- Built core platform features including authentication, role-based dashboards, search & filtering, interactive Google Maps, user submissions, Redis caching, Cloudinary image management, and Zod validation to improve usability and performance.
- Deployed on Vercel with rate limiting, firewall protection, performance monitoring, and comprehensive documentation, creating a scalable and secure production application.

RAG-based Knowledge Retrieval App | Sep 2025

- Engineered a Retrieval-Augmented Generation (RAG) system using Python, LangChain, OpenAI APIs, and ChromaDB, enabling semantic document retrieval and conversational question answering.
- Built a scalable retrieval pipeline including document ingestion, chunking, vector embeddings, indexing, and cosine similarity search to provide relevant contextual information for LLM inference.
- Optimized LLM response quality through prompt engineering, dynamic context injection, and retrieval-based grounding, improving factual accuracy and reducing hallucinations.

Silicon Valley Trail | Nov 2025

- Architected a modular Flask web application using the Application Factory pattern, Blueprints, SQLAlchemy, and SQLite to deliver a replayable simulation game with a scalable, maintainable backend.
- Engineered a rule-based game engine with persistent state management, event-driven workflows, and OpenWeather API integration with mock fallbacks, ensuring reliable gameplay despite external API failures.
- Applied software engineering best practices including service-layer architecture, unit testing with PyTest, error handling, environment-based configuration, and comprehensive documentation to improve maintainability and testability.

Experience

Software Engineer Fellow | 6/2024 – 5/2025

Springboard | Remote

- Built and deployed an end-to-end full-stack community platform enabling seamless access to legal, community, and educational resources by translating Figma designs into a fully functional application.
- Developed secure RESTful API for a dog adoption platform (Node.js, Express, MongoDB) with token authentication, 7 CRUD routes, MVC design, and Mocha/Chai tests, ensuring scalable data management.

- Engineered a Space Travel Planner app using React, Hooks and React Router with dynamic forms, client-side routing, React Testing Library delivering a seamless and fully tested user interface.

Tech Operations Associate | Nov/2021 – Oct/2022

Krishnamurthy Tandon Foundation | New York, NY

- Managed vendor relationships, coordinating deliverables, timelines, and third-party integrations, demonstrating project management, cross-functional collaboration, and process optimization.
- Streamlined digital systems and data organization, optimizing contact databases, records, and project documentation, to enhance efficiency, accuracy, and information accessibility.
- Led digital initiatives including social media management, ad campaigns, website updates, and technical setup for events and studio operations, showing problem-solving, adaptability, and hands-on technical skills.

Data Engineer Intern | 6/2019 – 8/2019

Real Factors (Acquired by 7Park Data) | New York, NY

- Resolved map display errors by correctly parsing geospatial data using SQL in BigQuery, ensuring accurate visualization before key acquisition deadline.
- Integrated and transformed multi-source data using SQL utilizing joins, partitions, aggregations and GeoJSON to create reliable tables and views for business intelligence and analytics.
- Developed interactive dashboard and BI tools in Apache Superset, providing actionable insights through charts, reports, while documenting processes for future team continuity.

Machine Learning Researcher (with Prof. Jia Xu) | 9/2018 – 5/2019

Hunter College | New York, NY

- Trained and optimized a Neural Attention Model (Codenn) using PyTorch on CUNY'S HPC, leveraging CUDA and Linux shell scripting to enhance performance and efficiency.
- Evaluated state-of-the-art NLP models for machine translation presenting findings to peers and professors to guide experimental design and implementation strategies.
- Documented insights and experimental outcomes, improving understanding of deep learning frameworks and contributing to group knowledge on HPC-based model optimization.

Education/Certification

Data Analytics, COOP Careers, 2026

Software Engineering Certification, Springboard, 2025

Bachelor of Arts in Computer Science, Hunter College, 2020

CUNY2X TTP Residency, Hunter College

Leadership

Board Member(Volunteer) | Lo-Nyamship Association USA, 2026 - 2028

President, School Student Alumni Association, 2026 - 2027